

Impact of Payroll on Major League Baseball Team Success

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INTRODUCTION

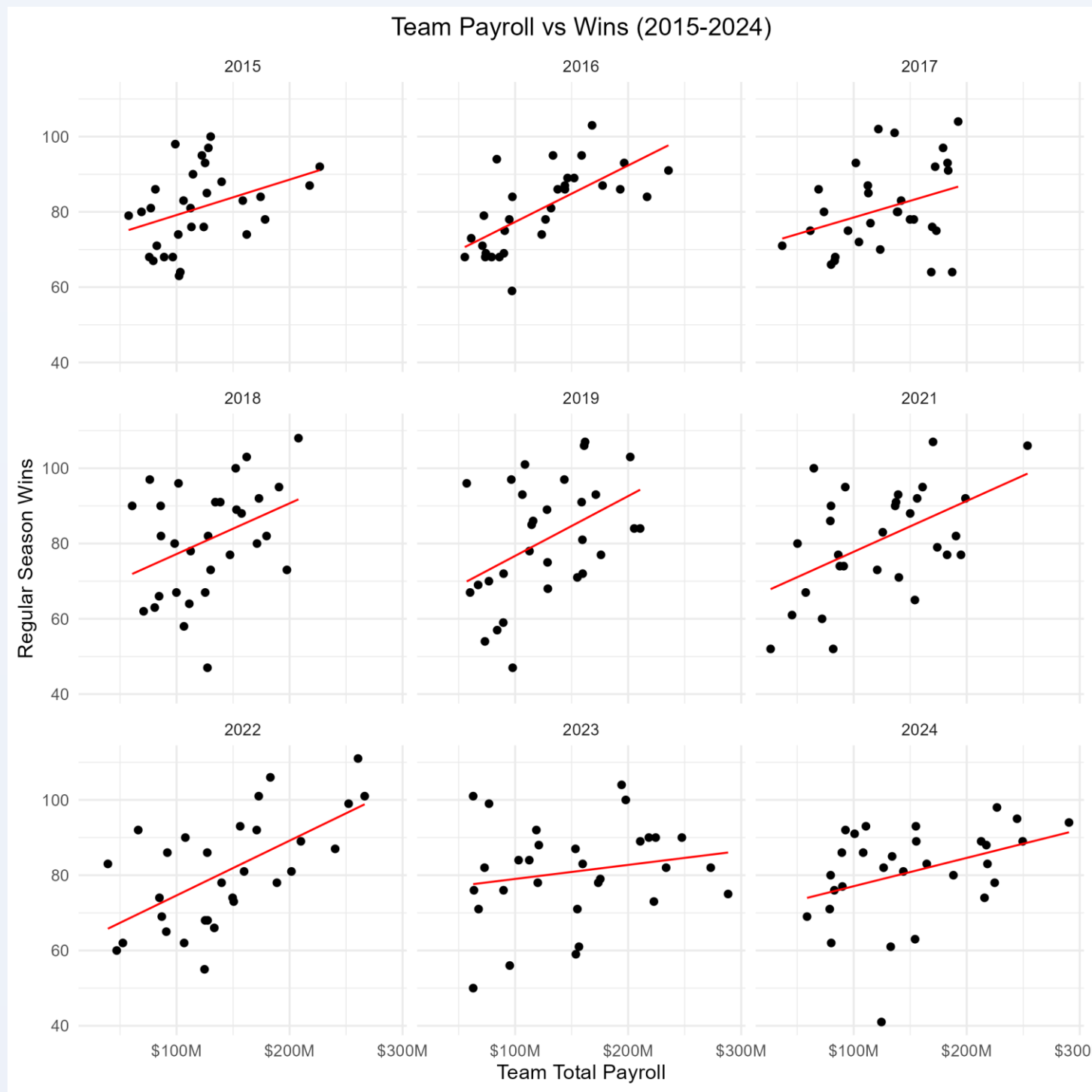
- Major League Baseball (MLB) is a game of strategy and one area of focus for teams is roster construction.
- A team's major league roster is constantly changing whether it be due to injuries, performance fluctuations, or financial constraints, making the task of managing a roster quite difficult.
- In recent seasons, record-breaking free agent contracts have raised questions about whether a team can simply buy their way to more wins.

RESEARCH QUESTION

- Can small-market teams achieve the same level of success as large-market teams despite payroll disparities?
- How does the distribution of payroll across different player categories impact a team's overall performance and year-to-year variation in wins?

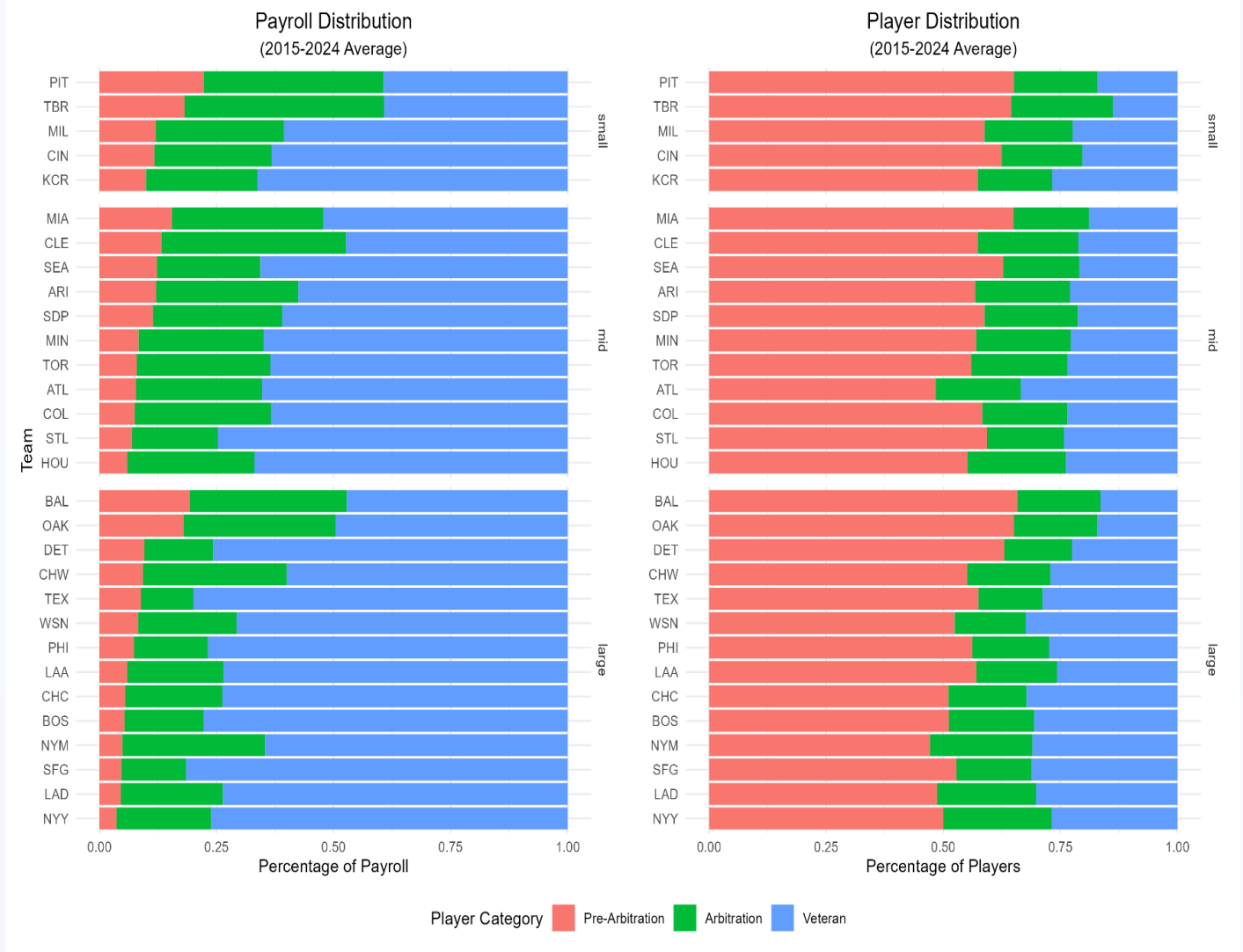
DATA

- Considered the period from 2015 – 2024. Excluded the COVID shorten 2020 season in analysis.
- Team performance data obtained from Baseball Reference.
- Active roster information obtained from Baseball Reference.
- Payroll and player status data were collected from Spotrac.
- Only included players that played in a least one game in a season for the team.
- Matched players from the Baseball Reference rosters with the Spotrac payroll information to create the dataset used for analysis.

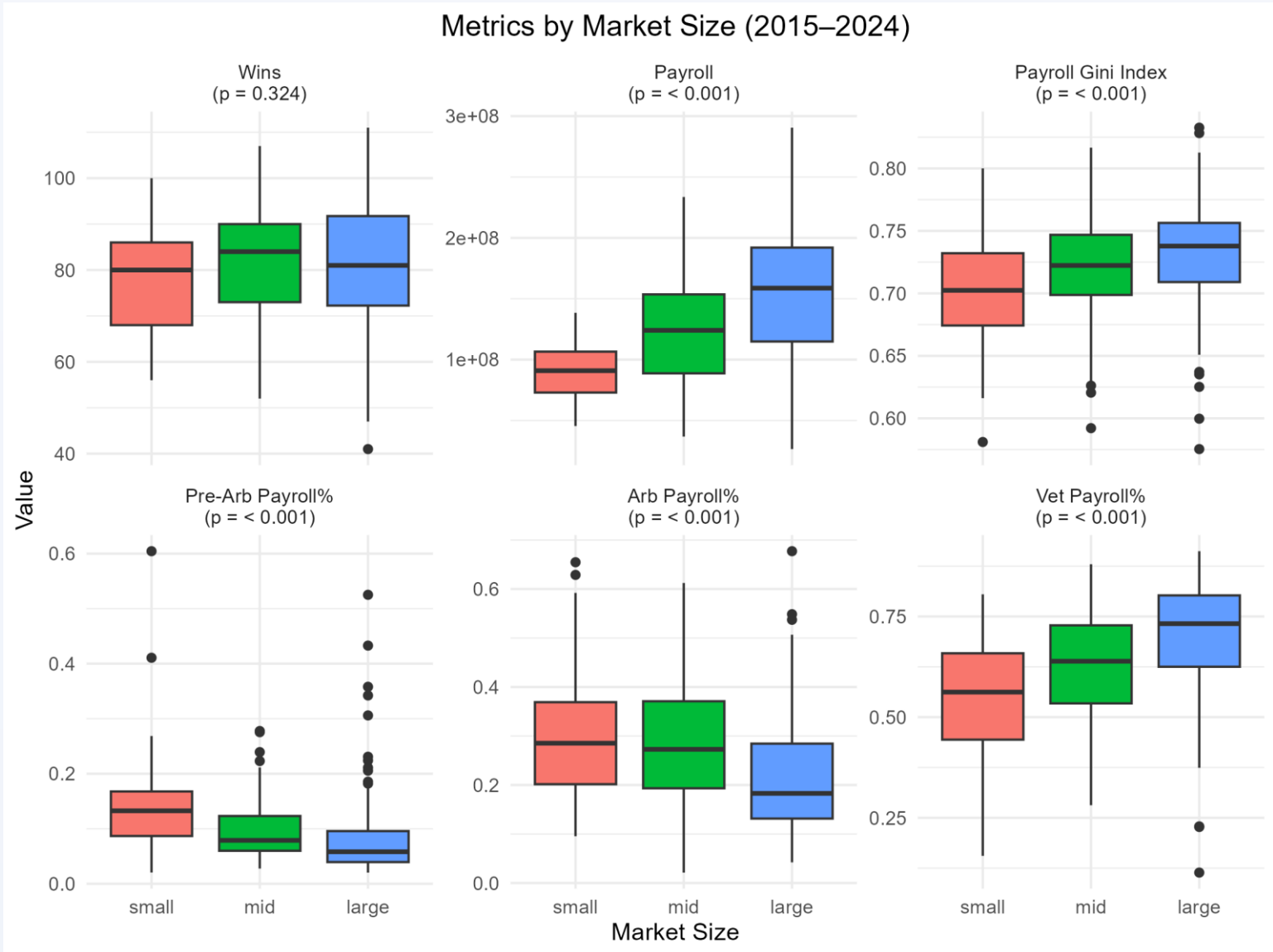


- In each season there is a positive correlation between a team's total payroll and their number of regular season wins.
- There has been a gradual increase in overall team payrolls, accompanied by a widening gap between the highest and lowest-spending teams.

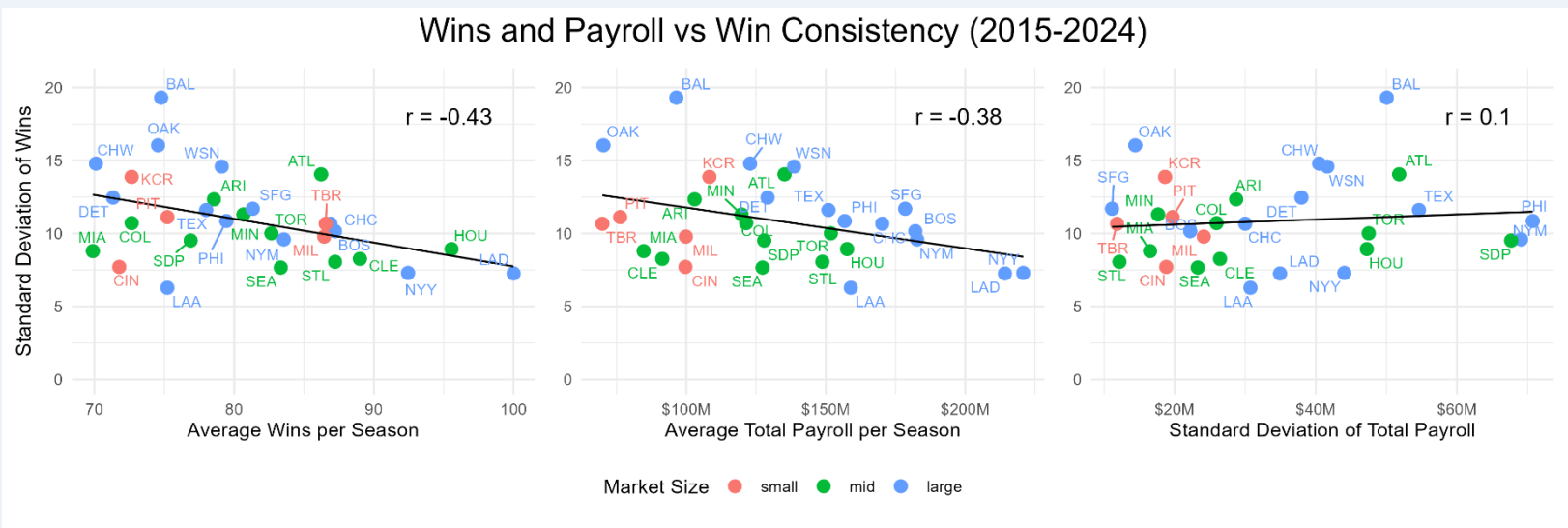
EXPLORATORY DATA ANALYSIS



- Pre-arbitration players make up a majority of each team's roster, but account for a minimal portion of the team's total payroll.
- The trends for payroll player category distribution are not uniform across the different market sizes.



- The ANOVA test shows that the difference in mean regular season wins are not statistically significant between market sizes.
- However, the other metrics have a significant difference suggesting a different strategy between market sizes.

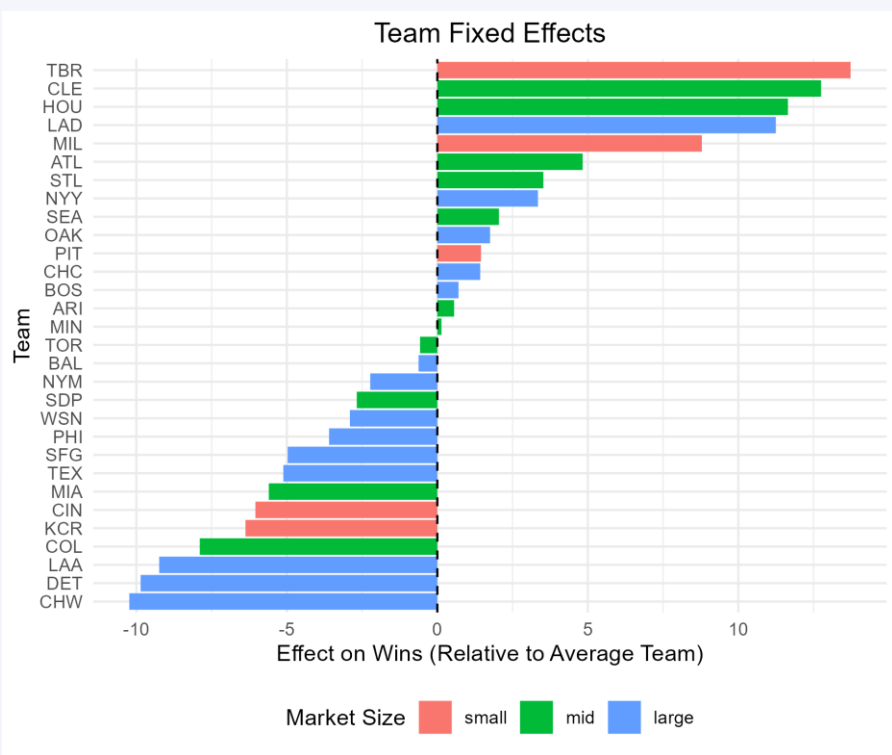


- Teams that win more games on average tend to have a more consistent performance.
- Teams with higher payrolls generally show slightly more consistency in performance.
- A majority of the large market teams show a higher variability in wins then their payroll would suggest.
- Teams with more variability in payroll each season had a slightly less consistent performance.
- Smaller market teams have less variability in payroll and lower total payroll but have an average win consistency.

REGRESSION RESULTS

Effect of Total Payroll on Regular Season Wins	Linear				Log			
	Baseline	Team FE	Year FE	Team + Year FE	Baseline	Team FE	Year FE	Team + Year FE
(Intercept)	67.532*** (1.960)				19.607* (8.302)			
total_payroll_m	0.101*** (0.014)	0.094*** (0.017)	0.107*** (0.017)	0.106*** (0.021)				
log(total_payroll_m)					12.764*** (1.720)	13.120*** (2.111)	13.240*** (1.609)	14.223*** (2.675)
Num.Obs.	270	270	270	270	270	270	270	270
R2	0.169	0.406	0.178	0.414	0.170	0.423	0.177	0.430
R2 Adj.	0.166	0.331	0.150	0.318	0.167	0.351	0.148	0.336
AIC	2106.1	2073.6	2119.0	2085.7	2105.6	2065.6	2119.6	2078.3
RMSE	11.87	10.04	11.80	9.96	11.86	9.89	11.81	9.83

- Across all models there is a significant positive relationship between team payroll and wins.
- Including team and year fixed effects improves the model fit. This suggests that team-specific factors explain a substantial portion of win variation.
- The log model suggest diminishing returns to payroll spending.



- All three market size categories have teams that both over- and underperform.
- TBR and CLE have developed systems that consistently outperform their financial limitations.

Effect of Payroll Allocations on Wins and Win Variability	Wins				Win Variability			
	Baseline	Total Payroll	Market Size	Payroll Interaction	Baseline	Total Payroll	Market Size	Payroll Interaction
pre_arb_payroll_pct	22.133* (8.896)	46.714*** (9.621)	21.822* (9.291)	50.559* (19.739)	14.720*** (3.152)	12.315*** (3.576)	16.769*** (3.257)	9.556 (7.335)
arb_payroll_pct	97.410*** (4.619)	86.538*** (4.838)	96.382*** (5.016)	90.276*** (12.236)	6.688*** (1.636)	7.752*** (1.798)	8.286*** (1.759)	6.884 (4.547)
vet_payroll_pct	83.488*** (1.825)	62.881*** (4.208)	83.250*** (1.864)	61.984*** (5.618)	8.162*** (0.647)	10.178*** (1.564)	8.292*** (0.654)	10.294*** (2.088)
total_payroll_m		0.102*** (0.019)		0.015 (0.301)		-0.010 (0.007)		0.045 (0.112)
market_size_small			0.242 (2.265)				-1.773* (0.794)	
market_size_mid			1.111 (1.686)				-1.096+ (0.591)	
arb_payroll_pct x total_payroll_m				0.059 (0.348)				-0.049 (0.129)
vet_payroll_pct x total_payroll_m				0.099 (0.301)				-0.058 (0.112)
Num.Obs.	270	270	270	270	270	270	270	270
R2	0.978	0.978	0.978	0.981	0.796	0.798	0.801	0.798
R2 Adj.	0.978	0.980	0.978	0.980	0.794	0.795	0.797	0.794
AIC	2118.7	2092.8	2122.2	2096.5	1558.4	1558.4	1556.2	1562.0
RMSE	12.06	11.45	12.05	11.45	4.27	4.26	4.22	4.25

- Across all models, how teams allocate their payroll strongly impacts wins and win consistency.
- Adding total payroll significantly improves the win model, but not the win variability model.
- Market size doesn't significantly affect team performance in an individual season once payroll structure is accounted for, but it does have a significant impact on win variability.
- The relationship between payroll allocation and wins as well as win variability doesn't significantly change based on total payroll size.

CONCLUSION

- Smaller market teams can achieve the same level of success as larger market teams, but they tend to take different approaches to roster construction.
- Teams with higher total payroll tend to find more regular season success. However, there is a diminished effect for higher spending teams.
- Team-specific factors account for a substantial portion of win variation once payroll is accounted for, indicating that effective management and strategy can overcome financial limitations.
- Arbitration players provide the best return on investment for wins.
- Pre-arbitration players contribute least to wins and cause more win variability
- Veterans contribute significantly to an increase wins but are also associated with higher win variability.
- While payroll matters in baseball, teams with financial limitation can compete effectively through strategic payroll allocation and strong organizational systems.

FUTURE DIRECTIONS

- This project focuses on regular season success, but this does not consider how teams do in the postseason. Similar analysis could be done on team's postseason wins.
- This project solely focuses on payroll and the distribution of payroll, but future work could consider individual player's performance as well as their position and how it relates to payroll and a team's overall success.
- This project also only considers the payroll for players who appeared in at least one game for the team and is unable to capture all the roster fluctuations across a season.

KEY TERMS

- Pre-Arbitration Player:** Has less than 3 years of Major League service time who is not yet eligible for salary arbitration; typically earns close to the league minimum salary.
- Arbitration-Eligible Player:** Has between three and six years of Major League service time (or "Super Two" status) who can negotiate salary through the arbitration process if no agreement is reached with the team.
- Veteran Player:** Has six or more years of Major League service time, making them eligible for free agency and usually commanding higher salaries.
- Market Size:** A classification based on a team's media market and revenue potential; grouped as small, mid-, or large-market.

ACKNOWLEDGEMENTS

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